# Variable subsumption and Logical Phonology

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**Abstract**

We examine the interaction of two phonological notions: Greek letter variables, ($\alpha$-notation), and natural classes in the context of Logical Phonology. The conclusions we reach hold for any theory that attempts to develop a precise mapping between phonological notation and mental processes/representations. By keeping the Greek variables in the grammar, rather than treating them as metalanguage variables, we predict where learners will generalize rules. Our "variable subsumption" model is a theory of what counts as a natural class.

This squib examines the interaction of two phonological notions: Greek letter variables, also known as $\alpha$-notation, and natural classes. We explore these ideas through the lens of Logical Phonology (e.g., Bale and Reiss 2018; Reiss 2021), henceforth LP. The conclusions we reach, however, hold for any theory that attempts to develop a precise mapping between phonological notation and mental processes/representations. LP seeks to provide a rigorous semantics that "interprets" phonological rules as string-to-string functions, with the goal of uncovering and addressing vagueness, ambiguities and inconsistencies in phonological theory. Greek letter variables represent a case in point. This commonly used notation threatens to undermine the intuitive notion of what constitutes a natural class if proper care isn't taken to ensure a coherent interpretation that can apply to any potential phonological rule.

We adopt a very simple view of phonological representation, laid out in Bale and Reiss (2018).

(1) Basic representational assumptions

a. Segments, like /p/ or /o/, are *sets* of valued features. For example, /o/ abbreviates the set {−HIGH, −LOW, +BACK, +ATR, +ROUND, . . . }.

b. Phonological representations are strings of segments. The set of valued features corresponding to a segment is associated with a timing slot, and these timing slots are ordered by precedence in lexical entries, as well as in words derived by morpheme concatenation.

c. Phonological rules represent mental functions that take strings of segments as inputs and produce strings of segments as outputs.

d. Greek letter variables are in the grammar—they are not a metalanguage device for expressing rule schemata (*pace* McCawley 1999).

In what follows, we explore some of the consequences of these assumptions.

Once we make explicit the idea that segments correspond to sets, natural classes can be defined as sets of segments, thus, sets of sets of valued features. For example, the set of high vowels in a language is the set of segments that are supersets of the set of features {+SYLLABIC, +HIGH}. For English, this natural class includes /i, ɪ, ɨ, u, ʊ/. In LP, curly brackets are used to denote sets of valued features, which may correspond to a particular segment, whereas square brackets are used to denote natural classes. For example, the expression in square brackets in (2) denotes the set of sets of valued features that are supersets of the set {+SYLLABIC, +HIGH} (i.e., the natural class of high vowels).

(2) A set of segments

$$\begin{bmatrix} +\textsc{Syllabic} \\ +\textsc{High} \end{bmatrix} = \left\{ x : x \supseteq \{+\textsc{Syllabic}, +\textsc{High}\} \right\}$$

By contrast, the expression in (3) with the curly brackets denotes a set of features, which could either represent an underspecified segment (e.g., a high vowel underspecified for whether it is + or − BACK, + or − LOW, etc.) or features that are added or subtracted in a phonological rule.

(3) A set of features

$$\begin{Bmatrix} +\textsc{Syllabic} \\ +\textsc{High} \end{Bmatrix}$$

Natural classes in phonology are examples of *partial descriptions*. A partial description defines a subset of the entities in a domain by characterizing a structure that is no more specified than any of the entities in that subset. For example, the vowels /i/ and /u/ are sets containing the elements −BACK and +BACK, respectively, but the natural class characterization in (2) makes no mention of BACK. In the case of individual segments, the set of features listed in the natural class is a subset of the set of features in each of the entities belonging to the class. This relationship is an example of a more general notion called subsumption.

(4) An information structure $A$ subsumes an information structure $B$ if all the information in $A$ is in $B$—all the pieces of $A$ are in $B$ and all the (relevant) relationships among pieces in $A$ hold in $B$ as well.

Subsumption is also used in phonology to express classes of strings of segments. For example, a rule that maps *si* to *ši* affects every string subsumed by *si*, so *asi, kasi, kosi, masipaka*, etc.[1]

Within phonology, the idea of subsumption—and thus natural class—is critical for understanding and interpreting phonological rules. Both the target and environment of a rule are defined by partial descriptions, usually in the form of natural classes. Rule application is implemented via subsumption. Consider the rule in (5) using the LP bracket convention.

(5) [−SONORANT] →{+VOICE} / __ [−SONORANT, +VOICE]

This rule says that any obstruent which occurs before a voiced obstruent becomes voiced. As pointed out in LP, *becoming voiced* is ambiguous. It either requires the deletion of −VOICED which is then replaced with +VOICE, or the addition of +VOICE if the target has no voicing feature. To simplify the discussion, we analyze → as two operations (see Bale et al. 2014), one that deletes the relevant feature(s) with regards to the sound change (e.g., removing −VOICE) and another that adds the new feature(s) in the sound change (e.g., adding +VOICE). For the rest of this squib, we focus on rules which add a feature, or set of features, to a segment (although nothing critical depends on this choice). We use the unification symbol ⊔ to represent this process, as in (6).

(6) [−SONORANT] ⊔ {+VOICE} / __ [−SONORANT, +VOICE]

[1] Because we are restricting discussion to segments and strings in this squib, we do not characterize subsumption for more complex structures, like syllables and feet. In other words, the only relevant relationship we discuss is precedence—and for the sake of simplicity, only immediate precedence. See Chapter 3 of Shieber (1986) for discussion of information structures and subsumption. Other useful sources are Sag et al. (1985) and Johnson (1988), as well as Strother-Garcia et al. (2017), which is specifically concerned with "substructures" in phonology.

This rule unifies {+VOICE} with any obstruent that occurs before a voiced obstruent.[2] A more formal interpretation of this rule is given in (7), where (6) is interpreted as a string-to-string (mental) function.

(7) The interpretation of (6) is the function from string $x_1 \ldots x_n$ to string $y_1 \ldots y_n$ such that if $x_i \in$ [−SONORANT], and $x_{i+1} \in$ [−SONORANT, +VOICE], then $y_i = x_i \sqcup \{+\text{ VOICE}\}$. Otherwise, $y_i = x_i$.

Such an interpretation strategy can be extended to any similar rule that fits the same schema. For example, let P, Q, and R be arbitrary lists of valued features:

(8) The interpretation of '[P] ⊔ {Q} / ___[R]' is the function from string $x_1 \ldots x_n$ to string $y_1 \ldots y_n$ such that if $x_i \in$ [P], and $x_{i+1} \in$ [R], then $y_i = x_i \sqcup \{\text{Q}\}$. Otherwise, $y_i = x_i$.

The resulting function essentially replaces segments $x_i$ that are members of the natural class [P] and that occur before members of the natural class [R] with the segment $x_i \sqcup \{\text{Q}\}$ (the segment unified with {Q}). Such an interpretation strategy works perfectly if P, Q, and R only contain valued features, however things become more complicated when Greek letter variables are included. For example, consider the rule in (9).

(9) [−SONORANT] ⊔ {$\alpha$VOICE} / ___ [−SONORANT, $\alpha$VOICE]

This rule is intended to say that an obstruent unifies with the set containing the valued feature for VOICE found on a following obstruent. As in (6), the target class includes any segment that is a superset of {−SONORANT}. However, the environment is defined by the partial description [−SONORANT, $\alpha$VOICE], and there is no segment that is a superset of the set {−SONORANT, $\alpha$VOICE}, because no segment literally contains the specification $\alpha$VOICE. All segments are either unspecified for VOICE, specified for +VOICE, or specified for −VOICE. The idea of natural classes defined by partial descriptions that subsume particular cases seems to fall apart. The interpretation schema in (8) fails because no segment $x_i$ is an element of [−SONORANT, $\alpha$VOICE], and even if there were such a segment, the resulting replacement should not contain $\alpha$VOICE.

Earlier work in generative phonology (e.g., Chomsky and Halle 1968; McCawley 1999) avoids this issue by treating Greek letter variables not as part of

[2] Unification is similar to union except that it does not permit unions which result in an inconsistent list of features. A list of features is inconsistent if it contains both +F and −F for any feature F.

grammar, but rather as a part of the phonologist's metalanguage. So an expression like (9) is taken to be an abbreviation for the set of rules in (10).

(10) i. $[-\text{SONORANT}] \sqcup \{+\text{VOICE}\}$ / ___ $[-\text{SONORANT}, +\text{VOICE}]$

ii. $[-\text{SONORANT}] \sqcup \{-\text{VOICE}\}$ / ___ $[-\text{SONORANT}, -\text{VOICE}]$

Since each rule in (10) does not contain any Greek letters, it can be interpreted via the general strategy in (8), with some extrinsic ordering of the two rules in (10).

Under such a treatment, an expression with only one Greek letter variable, as in (9), would correspond to two rules. However, as the number of Greek letter variables increases, the number of rules in the grammar increases exponentially by powers of 2, since each variable has two possible values. For example, rule (11) adds to (9) the condition that the target segment and the segment in the environment agree with respect to the feature CONTINUANT. This formula would lead to voicing assimilation between two stops or between two fricatives, but not between a stop and a fricative.

(11) $$\begin{bmatrix} -\text{SONORANT} \\ \alpha_1\text{CONTINUANT} \end{bmatrix} \sqcup \left\{\alpha_2\text{VOICE}\right\} \;/\; \_\_ \begin{bmatrix} -\text{SONORANT} \\ \alpha_1\text{CONTINUANT} \\ \alpha_2\text{VOICE} \end{bmatrix}$$

Understood as phonological shorthand, this expression corresponds to four rules, given the two values that each $\alpha$ can take. If rules need to be ordered, then the expression encodes four factorial (4! = 24) possible grammars.

If an expression mentions identity relations among, say fifteen features, then under this view, the Greek letter shorthand would correspond to $2^{15}$, or about 32,000 rules, which could represent, given rule ordering, more than $10^{35663}$ grammars. This is not outlandish to consider since many rules make reference to identical consonants, and the set of consonant features can be reasonably assumed to be at least 15.[3] In other words, an 'antigemination rule' (McCarthy, 1986; Odden, 1988; Reiss, 2003) that inserts a vowel between identical consonants is, under this view, a rule schema for thousands of individual rules, each of which must be individually listed in the grammar—that is what it means to *not* put the variables in

[3]For example, VOICE, NASAL, LATERAL, SONORANT, CONTINUANT, LABIAL, CORONAL, ANTERIOR, DORSAL, GLOTTAL, DELAYEDRELEASE, SPREADGLOTTIS, STRIDENT... This is a conservative estimate since, e.g., (Hayes, 2009, p. 95) uses twenty features to characterize consonants made with a single place of articulation and without taking into account ejectives, implosives and other phonation types.

the grammar.[4]

An alternative to treating Greek letters as metalanguage variables is to keep them in the grammar. This is far from trivial. It requires altering the interpretation schema in (8) to accommodate Greek letters into a more general theory of subsumption. To do so, we first need to define a set of variable assignment functions.

(12) **Set of Variable Assignment Functions**: For an arbitrary $n$, let $\mathcal{K}$ be the set of all functions $k$ whose domain is $\{\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_n\}$ and whose range is $\{-, +\}$.

An example of $\mathcal{K}$ is given in (13), where $n = 2$.[5] (An $n$ of 2 was chosen to save space. In practice, the $n$ would be much larger.)

(13)
$$k_1\text{:}\ \begin{array}{lll}\alpha_1 & \longrightarrow & + \\ \alpha_2 & \longrightarrow & +\end{array} \quad k_2\text{:}\ \begin{array}{lll}\alpha_1 & \longrightarrow & + \\ \alpha_2 & \longrightarrow & -\end{array} \quad k_3\text{:}\ \begin{array}{lll}\alpha_1 & \longrightarrow & - \\ \alpha_2 & \longrightarrow & +\end{array} \quad k_4\text{:}\ \begin{array}{lll}\alpha_1 & \longrightarrow & - \\ \alpha_2 & \longrightarrow & -\end{array}$$

With such a set, we can now interpret any list of features relative to one of these variable assignment functions.

(14) For any $k \in \mathcal{K}$

a. For any feature $\alpha_i$F where $\alpha_i$ is in the domain of $k$, $(\alpha_i\text{F})^k = k(\alpha_i)\text{F}$

b. For any feature +F and any feature −F, $(+\text{F})^k = +\text{F}$ and $(-\text{F})^k = -\text{F}$.

c. For any list of coefficient-feature pairs $f_1, f_2, \ldots, f_j$,
$$(f_1, f_2, \ldots, f_j)^k = (f_1)^k, (f_2)^k, \ldots, (f_j)^k.$$

Thus, for a list of features P = +VOICE, −NASAL, $\alpha_1$CONTINUANT, $\alpha_2$LABIAL, the intepretation of $(\text{P})^{k_1}$ would equal +VOICE, −NASAL, +CONTINUANT, +LABIAL whereas $(\text{P})^{k_2}$ would equal +VOICE, −NASAL, +CONTINUANT, −LABIAL.

With this set of variable assignment functions in mind, we can now generalize what it means for a particular segment to be subsumed by a "natural class" with a Greek letter variable. We call this *variable subsumption*. There are two key ideas: 1) a segment is subsumed by the "natural class" relative to a particular variable assignment and 2) the same variable assignment that allows for variable

[4] Reiss (2003) argues that the existence of non-identity conditions on rules demonstrates that autosegmental linking is insufficient to express the full range of antigemination and anti-antigemination phenomena and that the equivalent of quantificational logic is needed in the phonology. The point is that these thousands of rules cannot be replaced by a single statement making use of PLACE nodes linked to seperate timing slots.

[5] By $\mathcal{K}$ we mean $\mathcal{K}_n$ where for each $k \in \mathcal{K}_n$ and any $i \leq n$, $\alpha_i$ is in the domain of $k$. For readability, we omit subscripts when their value is obvious or inconsequential.

subsumption, or at least one that establishes the same subsumption relationship, is also used to implement the segment substitution. In (15), we do this by constructing a set $\mathcal{K}'$ which consists of all and only those variable assignments where variable subsumption is possible—i.e., where the target segment is a member of the target natural class under this variable assignment and where the triggering segment (the segment to the right) is a member of the triggering natural class under this variable assignment.

(15) The interpretation of '[P] ⊔ {Q} / __[R]' is the function from string $x_1 \ldots x_n$ to string $y_1 \ldots y_n$ such that if the set $\mathcal{K}' = \{k \in \mathcal{K}\text{: } x_i \in [(\mathrm{P})^k]$ and $x_{i+1} \in [(\mathrm{R})^k]\}$ is empty, then $y_i = x_i$. Otherwise, $y_i = x_i \sqcup \{(\mathrm{Q})^k\}$, where $k$ is an arbitrarily chosen member of $\mathcal{K}'$.

If $\mathcal{K}'$ is empty, then subsumption is impossible, either due to the fact that the target segment or triggering segment does not contain a specified +F or −F, or due to the fact that there is no assignment for a Greek letter variable that permits both the target and triggering segment to be members of their respective natural classes. An empty $\mathcal{K}'$ thus leads to an identity mapping (i.e., $y_i = x_i$).

However, if $\mathcal{K}'$ is not empty, then there is an assignment $k$ that permits subsumption of both the target and triggering segment. We can then use this assignment, or any of the others, to implement the featural change in the target segment. The choice of $k$ does not matter. No matter which $k$ is chosen from $\mathcal{K}'$, the output is the same—i.e., for any $k'$ and $k''$ that are members of $\mathcal{K}'$, $(\mathrm{Q})^{k'} = (\mathrm{Q})^{k''}$. A proof is given in (16).

(16) Let $x_i$ and $x_{i+1}$ be (consistent) segments. For all lists of features P, Q, and R, and for any $k'$ and $k''$ that are members of $\mathcal{K}$, if $k'$ and $k''$ are both members of $\mathcal{K}' = \{k \in \mathcal{K}\text{: } x_i \in [(\mathrm{P})^k]$ and $x_{i+1} \in [(\mathrm{R})^k]\}$ and if any variable in the list Q is also in either P or R, then $(\mathrm{Q})^{k'} = (\mathrm{Q})^{k''}$.

*Proof.* Suppose that $k'$ and $k''$ are both members of $\mathcal{K}'$ and that any variable in the list Q is also in either P or R. For the purpose of contradiction, assume that $(\mathrm{Q})^{k'} \neq (\mathrm{Q})^{k''}$. Given that $(\mathrm{Q})^{k'} \neq (\mathrm{Q})^{k''}$, there must be a $\alpha_j$ in Q, such that $k'(\alpha_j) \neq k''(\alpha_j)$ (i.e., either $k'(\alpha) = +$ and $k''(\alpha_j) = -$, or $k'(\alpha_j) = -$ and $k''(\alpha_j) = +$). By assumption, either $\alpha_j$F is in P, for some feature F, or $\alpha_j$G is in R, for some feature G. Since both $k'$ and $k''$ are in $\mathcal{K}'$, it follows that either $\alpha_j$F is in P and both +F and −F are members of $x_i$ or $\alpha_j$G is in R and both +G and −G are members of $x_{i+1}$. Either way, given that $x_i$ and $x_{i+1}$ are consistent segments, we reach a contradiction. Thus, $(\mathrm{Q})^{k'} = (\mathrm{Q})^{k''}$. □

Critically, the interpretation in (15) is equivalent to the one in (8) if there are no Greek letter variables. In such cases, when subsumption holds $\mathcal{K}' = \mathcal{K}$. The choice of $k$ will not matter since $k$ cannot affect the interpretation of any list of features that does not contain Greek letters. In other words, if P and R contain no Greek variables, then they correspond to traditional natural class representations—they denote the class of all and only segments that are supersets of the set of features listed in P and R, respectively.

We explored the interaction between subsumption and Greek letter variable by examining rules of the form '[P] $\sqcup$ {Q} / __[R]', but the general strategy expands trivially to other schemas as well, such as those in (17).

(17)
a. [P] $\sqcup$ {Q} / [R]__
b. [P] $\sqcup$ {Q} / [R]__[T]
c. [P] $\rightarrow${Q} / __[R]
d. [P] $\rightarrow${Q} / [R]__
e. [P] $\rightarrow${Q} / [R]__[T]

Notice that the combinatoric explosion of rules that accompanies the idea of Greek letter variables as metalanguage entities is avoided under the approach advocated here. A given input string must be checked to see whether in each rule position (target and environment segment classes) values match for each feature with an $\alpha$ value in the rule. In the worst case scenario, this involves checking three positions (P, R, T) for as many features as there are, say, 25 or so. This appears to be much more efficient computationally than cycling through thousands of rules that *must* be extensionally listed in the grammar under the view that Greek variables are *not* in the grammar. By keeping the Greek variables in the grammar we predict where learners will generalize rules beyond specifically encountered combinations of feature coefficients—variable subsumption is a theory of what counts as a natural class.

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